

# NUMERICAL ANALYSIS

## PROBLEM SET 1

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We begin by noting that all of the files referenced in this document are available at:  
<https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/ProblemSet1>

1. The value of  $\pi$  is given by the alternating series:

$$\pi = 4 \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = 4 \left( 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right)$$

(a) Use this series to construct an algorithm to approximate the value of  $\pi$  to within a specified tolerance,  $\varepsilon$ . The algorithm should be general to allow for any  $\varepsilon$ . Provide a theoretical upperbound  $\epsilon$  on the error of your approximation. Hint. It may be useful to recall known properties of the partial sums of alternating series.

(b) Run your algorithm with a tolerance value of  $\varepsilon = 5 \times 10^{-3}$  and show that your result is within  $\varepsilon$  of  $\pi = 3.1415926 \dots$

(c) Make a plot of  $\varepsilon$  versus the number  $N$  of required terms in your algorithm on a log-log scale. What is the slope of the line that the resulting plot approaches as  $N$  increases? Plot the actual error as a function of  $N$  on the same axes, you may treat the stored value of  $\pi$  as “exact.” How does the actual error compare to the estimated threshold?

*Solution.* (a) Our algorithm is given in the file “problem1.m” but also below as it is simple:

```
function approximatePi = computePiPlotError(error)
    p = 0;
    sum = 0;
    index = 0;
    pApproximated = [];
    indexedError = [];
    while (abs(pi - p) > error)
        sum = sum + (-1)^index * 1/(2*index + 1);
        p = 4*sum;
        index = index + 1;
        pApproximated(index) = p;
        err = abs(pi - p);
```

```

        indexedError(index) = err;
    end
    approximatePi = p;
    loglog(1:index,indexedError,'--k')
    xlabel('Iteration Index n');
    ylabel('Error');
    title('Error in Pi Approximation');
end

format long
approx_pi = computePiPlotError(5e-3)
pi - approx_pi

```

Note, an upper bound on  $\varepsilon$  should be  $|pi - 4| \approx 0.85840$ .

(b) When we run this function we get the following output from matlab:

$$p = 3.136592684838816$$

$$\varepsilon = 0.004999968750977$$

(c) When we plot  $\varepsilon$  versus the number of iterations of our algorithm to try to find  $\pi$  we get: I'm not exactly sure what we're going for here with the "actual error". The only thing

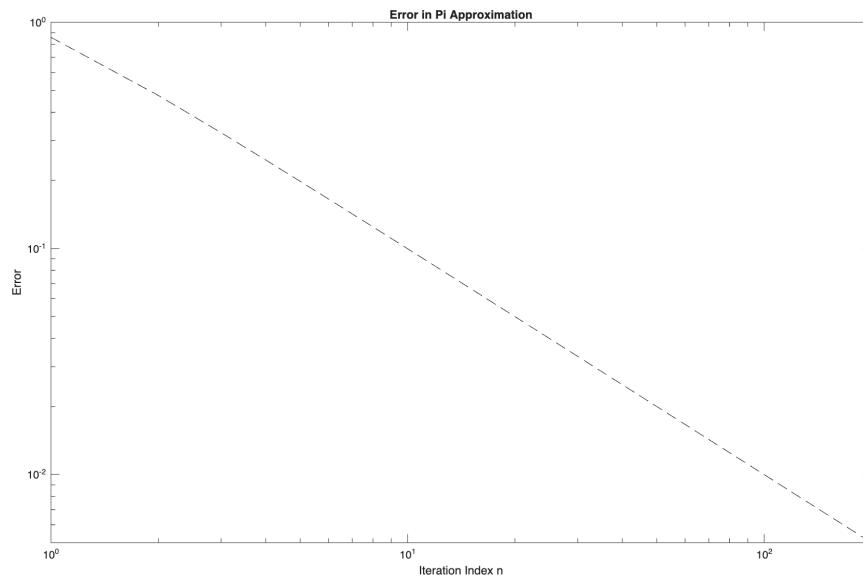


FIGURE 1. Log-Log plot  $\varepsilon$  versus  $N$

that I can come up with is considering the remainder of the tail of the summation after the

$n^{\text{th}}$  iteration.

$$\text{err}_n^{\text{actual}} = 4 \sum_{k=n+1}^{\infty} \frac{(-1)^k}{2k+1}$$

I suppose we could pick some huge  $k$ , say  $10^{12}$  and iterate from our current  $n$  to there to get the actual error. But to do so I would end up using the symbolic computation capabilities in matlab. I've added a "problem1b.m" in the directory but the computational capacity on the "symsum" function is insufficient for the problem.

□

**2.** Numerically determine which of the following limits approaches 1 faster. Then confirm the numerical evidence by determining the rate of convergences of each sequence

$$\lim_{n \rightarrow 0} \frac{\sin(x^2)}{x^2} \quad \text{versus} \quad \lim_{n \rightarrow 0} \frac{(\sin x)^2}{x^2}.$$

*Solution.* We'll just run a table here for  $\{10^0, 10^{-1}, 10^{-2}, \dots, 10^{-9}\}$  to look at convergence of the functions listed above. The code for this problem is minimal command line work using vectors with the scalar versions of the sin and arithmetic functions.

| $n$       | $\sin(x^2)/x^2$    | $(\sin x)^2/x^2$  |
|-----------|--------------------|-------------------|
| $10^0$    | -0.005063656411098 | 0.002959589690933 |
| $10^{-1}$ | 0.841470984807897  | 0.708073418273571 |
| $10^{-2}$ | 0.999983333416666  | 0.996671107937918 |
| $10^{-3}$ | 0.999999998333333  | 0.999966667111108 |
| $10^{-4}$ | 0.99999999999833   | 0.99999666666711  |
| $10^{-5}$ | 1.000000000000000  | 0.99999996666667  |
| $10^{-6}$ | 1.000000000000000  | 0.99999999966667  |
| $10^{-7}$ | 1.000000000000000  | 0.9999999999667   |
| $10^{-8}$ | 1.000000000000000  | 0.9999999999997   |
| $10^{-9}$ | 1.000000000000000  | 1.000000000000000 |

Now as to find the rate of convergence for the functions, we'll consider them one at a time.

For  $\sin(x^2)/x^2$  notice that

$$\frac{\sin(x^2)}{x^2} \leq \frac{1}{x^2}$$

So our rate of convergence is  $\mathcal{O}(1/x^2)$ .

For  $(\sin x)^2/x^2$  we first notice that

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

Therefore, we see that

$$\lim_{x \rightarrow 0} \frac{(\sin x)^2}{x^2} = \left( \lim_{x \rightarrow 0} \frac{\sin(x)}{x} \right)^2 = 1^2 = 1$$

and our convergence is on the order of  $\mathcal{O}(1/x)$ .

□

**3.** (a) Consider the function  $f_c(x) = x^2 - 12x + c$ . What are the zeros of  $f_c$  with  $c = 35$ ? What are the zeros of  $f_c$  with the slight constant shift  $c = 35 - 10^{-2}$ ? Relative to the size of change in the constant term  $c$ , how big is the change in the zeros of the function?

(b) consider the function  $g_c(x) = x^2 - 10x + c$ . What are the zeros of  $g_c$  with  $c = 25$ ? What are the zeros of  $g_c$  with the slightest constant shift  $c = 25 - 10^{-2}$ ? Relative to the size of change in the constant term, how big is the change in the zeros of the function?

(c) Compare the nature of the roots of  $f$  and  $g$ . How might this contribute to the relative changes in the zeros for each function?

*Solution.* (a) Using the quadratic formula we quickly see that:

$$x_a = \frac{12 \pm \sqrt{144 - 4(35)}}{2} = \{7, 5\}$$

but if we let  $c = 35 - 10^2$ , our zeros become (using matlab for the calculation):

$$x_b = \frac{12 \pm \sqrt{144 - 4(35)}}{2} = \{7.048808848170152, 4.951191151829848\}$$

and we note that the values came from the opposing fomulae. The shift in the zero is  $x_a - x_b$  which follows as

$$x_a - x_b = \{0.048808848170152, 0.048808848170152\}.$$

(b) Similarly to (a) we use the quadratic formula to quickly see that we have a singular root at:

$$x_a = \frac{10 \pm \sqrt{100 - 4(25)}}{2} = 5$$

However, if we let  $c = 25 - 10^{-2}$ , we get the following:

$$x_b = \frac{10 \pm \sqrt{100 - 4(25 - 10^{-2})}}{2} = \{5.316227766016840, 4.683772233983160\}$$

The shift in zero is  $x_a - x_b = 0.316227766016840$ .

(c) The nature of the roots differ from  $f$  and  $g$  in that  $g$  is tangent to the  $x$  axis at 5, where as  $f$  has two distinct roots at  $c = 35$ , 5 and 7. □

**4.** Identify the potential roundoff error problems in the following algorithm for calculating the roots of the quadratic equation  $ax^2 + bx + c = 0$ .

**GIVEN:** real coefficients  $a, b, c$

STEP 1: calculate  $disc = \sqrt{b^2 - 4ac}$   
 STEP 2: calculate  $root1 = -2c/(b + disc)$   
 STEP 3: calculate  $root2 = (-b/a) - root1$   
 OUTPUT:  $root1$  and  $root2$

Note that this algorithm uses the fact that the sum of the roots of  $ax^2 + bx + c = 0$  is equal to  $-b/a$ .

*Solution.* If we're considering all the possibilities for roundoff error, we start by noticing that we initially have the possibility for roundoff error in that the coefficients  $a, b, c$  may in fact be irrational needing to be initially rounded.

In STEP 1 we can have rounding problems when  $b^2 \approx 4ac$  by subtracting numbers that are nearly identical.

In STEP 2 we can have real problems when  $b + disc \approx 0$  because the denominator has large relative error.  $root1$  in this case becomes highly inaccurate and may overflow. Also if  $c$  is tiny  $-2c$  could underflow.

In STEP 3 if we are dealing with subtraction of nearly equal numbers then we could have roundoff error. The cases when this could occur would be if the roots are of very different magnitudes and the large one is  $root1$ . Lastly notice that if  $a$  is tiny,  $-b/a$  can overflow.  $\square$

5. (a) Plot the function  $f(x) = \ln(1+x) - \cos(x) - x + 1$  over the interval  $-5 \times 10^{-6} \leq x \leq 5 \times 10^{-6}$  using 1001 uniformly spaced points in  $x$ .

(b) Reformulate  $f$  to avoid the cancellation error, and then repeat part (a).

(c) Describe the issues with the original formulation in part (a) and how they were resolved with the reformulation in part (b).

*Solution.* (a) Running the plot using this order of operations yields the plot in Figure 2.

(b) If we simply rearrange the function to put the additions and subtractions in a different order, for example

$$f(x) = (\ln(1+x) - x) + (1 - \cos(x))$$

we get the graph in Figure 3, which we overlayed on top of part (a).

(c) The main issue with the problem stemmed from roundoff error around the subtraction between  $\cos(x)$  and  $x$  because we're so close to 0. So that essentially forces us into buckets as we see in the plot in Figure 2.  $\square$

6. (a) Determine the third-degree Taylor polynomial associated remainder term for the function  $f(x) = \sqrt{1+x}$ . Use  $x_0 = 0$ .

(b) Using the result of part (a), approximate  $\sqrt{1.5}$ , and compute the theoretical error bound associated with this approximation. Compare the theoretical error bound with the actual error.

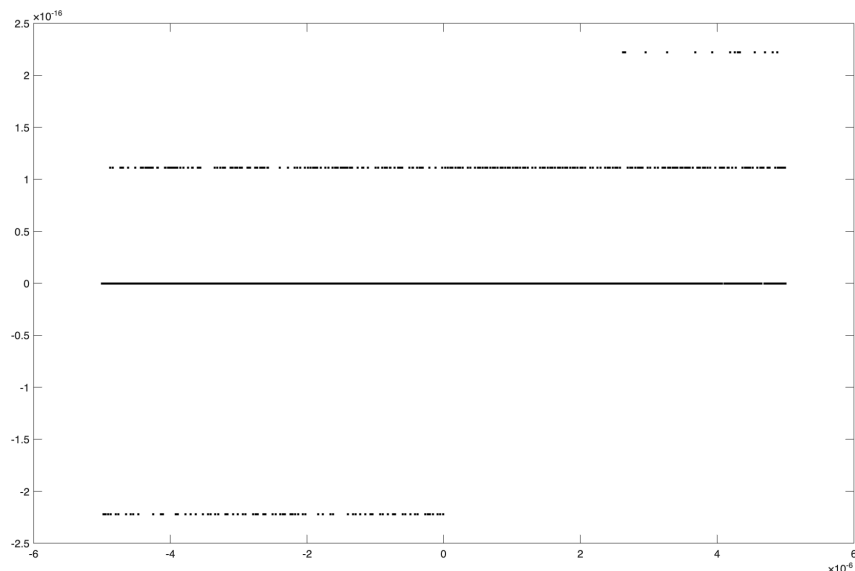


FIGURE 2. Plot of  $\ln(1+x) - \cos(x) - x + 1$  using 1001 uniformly spaced points over the range

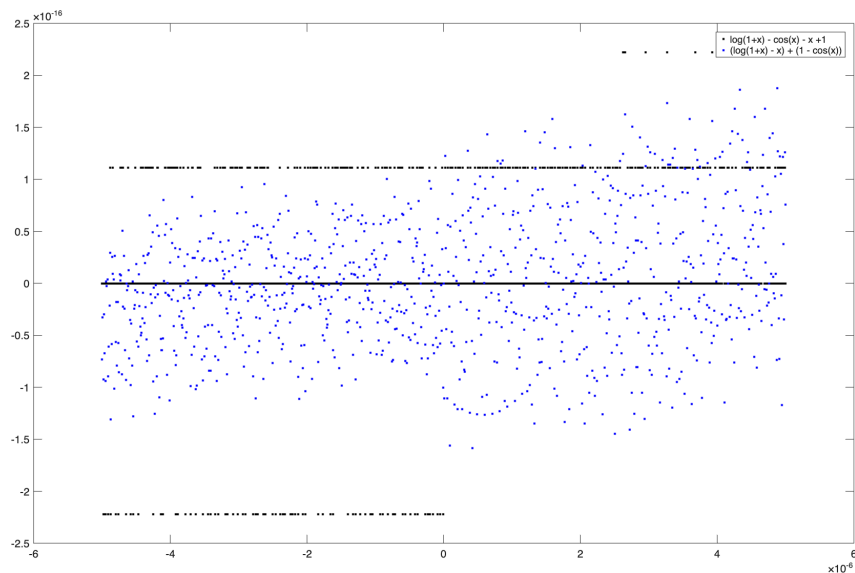


FIGURE 3. Plot of  $\ln(1+x) - \cos(x) - x + 1$  versus  $(\ln(1+x) - x) + (1 - \cos(x))$

(c) Compute the following limit and determine the corresponding rate of convergence:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - \frac{1}{2}x}{x^2}$$

*Solution.* (a) We know that for an arbitrary function  $f(x)$  the third degree Taylor polynomial at  $x_0 = a$  follows as:

$$T_3(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \mathcal{O}$$

and for our function  $f(x) = \sqrt{1+x}$  we have that:

$$\begin{aligned} f'(x) &= \frac{1}{2}(1+x)^{-1/2} \\ f''(x) &= -\frac{1}{4}(1+x)^{-3/2} \\ f'''(x) &= \frac{3}{8}(1+x)^{-5/2} \end{aligned}$$

So using  $x_0 = 0$  we get that

$$T_3(x) = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}$$

Note that the Lagrange remainder term follows as

$$R_3(x) = \frac{f^{(4)}(\xi)}{4!} = \frac{-\frac{15}{16}(1+\xi)^{-7/2}}{24}x^4 = -\frac{15x^4}{384(1+\xi)^{7/2}}.$$

(b) Approximate  $\sqrt{1.5} = \sqrt{1+1/2}$  as

$$\begin{aligned} T_3\left(\frac{1}{2}\right) &= 1 + \frac{1}{4} - \frac{1}{24} + \frac{1}{128} \\ &= 1 + 0.25 + 0.03125 + 0.0078125 \\ &= 1.2265625. \end{aligned}$$

For the theoretical error bound we notice

$$f^{(4)}(x) = \frac{15}{16}(1+x)^{-7/2} \leq \frac{15}{16}$$

So we have that:

$$|R_3\left(\frac{1}{2}\right)| \leq \frac{\max|f^{(4)}|}{4!} \left(\frac{1}{2}\right)^4 \leq \frac{15}{6144} \approx 0.00244140625$$

Whereas, the true value we get (from matlab) is  $\sqrt{1.5} = 1.224744871391589$  so the actual error is

$$|1.2265625 - 1.224744871391589| = 0.001817628608411$$

(c) If we use the Taylor series expansion of the numerator around  $x = 0$ , we get that

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \mathcal{O}(x^3)$$

which we plug in as the numerator, and we get

$$\sqrt{1+x} - 1 + \frac{1}{2}x = \left(1 + \frac{x}{2} - \frac{x^2}{8} + \mathcal{O}(x^3)\right) - 1 + \frac{1}{2}x = -\frac{1}{8}x^2 + \mathcal{O}(x^3)$$

So we see that

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 + \frac{1}{2}x}{x^2} = \lim_{x \rightarrow 0} \frac{-\frac{1}{8}x^2 + \mathcal{O}(x^3)}{x^2} = -\frac{1}{8}$$

And the rate of convergence follows as  $x^2$  since those are the significant terms in the limit.  $\square$

7. Consider the linear system  $Ax = b$  with  $A = \begin{bmatrix} 3.4 & 2.8 \\ 8.0 & 6.6 \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ , and  $b = \begin{bmatrix} 3.4 \\ 8.0 \end{bmatrix}$ .

(a) Determine the analytic solution by hand. (How do we know that it is unique, i.e. that the problem is well-posed?)

(b) Now solve the system with  $b = \begin{bmatrix} 3.41 \\ 8.0 \end{bmatrix}$ . You may use your computer.

(c) Given the difference in solutions for parts (a) and (b), comment on the conditioning of the problem. What is  $\det[A]$  and for what values with the problem be ill-posed?

*Solution.* (a) First we write out the system of equations as

$$\begin{bmatrix} 3.4 & 2.8 \\ 8.0 & 6.6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3.4 \\ 8.0 \end{bmatrix}$$

So I decided to go old school and use a plain system of equations and do substitution which looks like:

$$3.4x_1 + 2.8x_2 = 3.4$$

$$8.0x_1 + 6.6x_2 = 8.0$$

slight algebra yields  $x_1 = 1 - 2.8/3.4x_2$ . Plugging this into the second equation we get

$$\begin{aligned} 8.0 - \frac{8.0 \cdot 2.8}{3.4}x_2 + 6.6x_2 &= 8.0 \\ \frac{8.0 \cdot 2.8}{3.4}x_2 + 6.6x_2 &= 0 \end{aligned}$$

So  $x_2 = 0$  and  $x_1 = 1$ .

(b) Now we set the right hand side of our linear system of equations to be  $[3.418.0]^\top$ . Our equations fall out as

$$3.4x_1 + 2.8x_2 = 3.41$$

$$8.0x_1 + 6.6x_2 = 8.0$$

So we get that  $x_1 = 3.41/3.4 - 2.8/3.4x_2$ . Which if we substitute back in to our second equation yields

$$\begin{aligned} \frac{8.0 \cdot 3.41}{3.4} - \frac{8.0 \cdot 2.8}{3.4}x_2 + 6.6x_2 &= 8.0 \\ 8.023529411764706 - 6.588235294117647x_2 + 6.6x_2 &= 8.0 \\ 0.011764705882353x_2 &= -0.023529411764706 \\ x_2 &= -2 \end{aligned}$$

So we have that  $x_1 = -0.644117647058823$ .



(c) It's fairly odd to have such a widely different set of solutions for such a small change in our linear system of equations. But if we compute the determinant of the  $2 \times 2$  matrix, we get:

$$\det \begin{bmatrix} 3.4 & 2.8 \\ 8.0 & 6.6 \end{bmatrix} = 3.4 \cdot 6.6 - 8.0 \cdot 2.8 = 22.44 - 22.4 = 0.04.$$

which actually makes considerable sense as such a low determinant would yield an unstable equilibrium of solutions. Clearly if the  $\det[A] = 0$  the problem is ill-posed.  $\square$